

FieldTest Recorder

- **Problem Statement ID** – SIH26231
- **Problem Statement Title** – Digital Companion for Field Drug Testing
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – [Team ID]
- **Team Name** – [Team Name]



FieldTest Recorder

A guided, measured and signed field test for NCB's Narcotic Drugs Detection Kit

THE GAP TODAY

- **Read by eye.** Officers match the kit's colour against a printed chart; nothing proves a test happened at a place and time (NCB, this problem statement).
- **No standard.** Bombay HC (2021): NCB "has not prescribed the standards"; field testing is "arbitrary", and a bare mention in the panchnama is not enough.
- **Wrong drug, wrong quantity.** Kerala HC (2024, Anuraj): seized "MDMA" was methamphetamine, which changes the NDPS quantity category.
- **So:** disputed seizures, bail on procedure, and no data on how often field tests are wrong.

HOW IT WORKS (OFFLINE, ON THE OFFICER'S PHONE)

- 1 Guide**
NCB kit Tests A–E in the printed flow-chart order: drops, wait time, which layer to read
- 2 Capture**
Photo of the reaction on a reference colour card, next to a reagent-only blank well
- 3 Read**
Colour bands from the kit's own ranges: positive / no colour change / inconclusive + every drug it fits
- 4 Sign**
Time, GPS, officer and image hash, signed in phone hardware and hash-chained
- 5 Anchor**
A 14-character code written into the witness-signed panchnama
- 6 Report**
Field Test Memo, Form-1 item 5, BSA s.63 data; anyone can verify offline

WHAT IS NEW

Says only what the chemistry supports

Test E blue = "cocaine or methaqualone" until E3/E4 narrows it; warns when the NDPS quantity category can't be told in the field.

Holds up when the sample amount varies

SIMULATED

Bands + blank well: 0% false positives where single-point colour matching gave up to 11.7%.

Proves completeness, not just integrity

PROVEN IN CODE

Deleted or re-run tests are caught; not even our server admin can forge a record.

No new hardware

Existing NCB kit + officer's phone + a printed card. Works offline; embeds in e-Sakshya as an SDK.

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TECHNICAL APPROACH



OFFICER'S PHONE (offline SDK)

Guided NDDK protocol

signed: steps, drops, read times, flow charts

Capture

CameraX / Camera2 RAW, card located by ArUco markers

Colour engine

card correction, blank well, CIELAB colour bands

Record core

ECDSA P-256 in StrongBox/TEE, SHA-256 chain

Log + reports

searchable log, Field Test Memo, Form-1 item 5

sync when
online

THIN SERVER: NIC / MeitY cloud

Enrolment

Android key attestation + supervisor's DSC

Sync

idempotent, fork/replay checks, signed receipts

Storage

PostgreSQL + S3 write-once, durable checkpoints

Search + lab loop

case search; Rule 14 lab result vs field result

Never trusted

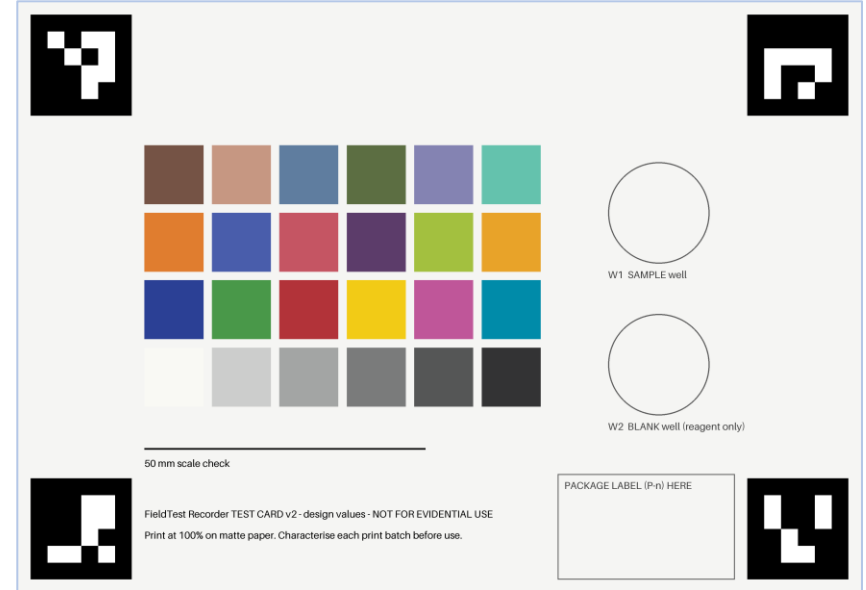
records verify without the server

Trust without trusting us: NCB signs the supervisor list; a supervisor's DSC binds officer + phone + key; the phone signs and chains every test; the chain heads of every phone go into the witness-signed panchnama. An offline verifier (same record core) checks it all in court.

PROVEN IN CODE

Stack: Kotlin · CameraX/Camera2 · OpenCV · Android Keystore + key attestation · SQLCipher · RFC 8785 JSON · ECDSA P-256 · Spring Boot · PostgreSQL · S3 object lock · Parichay / Class 3 DSC

Deliberately not used: blockchain, cloud AI, a neural-network "drug detector" (a colour test cannot identify a drug; the court must be able to re-run the reading)



Reference card v2 (printable, 150 x 105 mm): 24 colour patches, 4 markers, SAMPLE and BLANK zones. The engine finds it in the print file.

BUILT SO FAR

PROVEN IN CODE

Reference implementation: record core, verifier, colour engine, NDDK protocol, sync server, search

SIMULATED

17 experiments with 95% intervals (colour, liveness, night light, amount)

NEXT

Android app (Kotlin) on the same test vectors

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FEASIBILITY AND VIABILITY



18/18

PROVEN IN CODE

tampering attacks caught, incl. insider-admin forgery and deleted records

15/15

PROVEN IN CODE

sync and failure scenarios handled, incl. restore from an old backup

11.7% → 0%

SIMULATED

false positives when the sample amount varies (single-point rule vs colour bands)

89%

SIMULATED

NCB-kit chart outcomes read correctly, 1.7% wrong, rest inconclusive

CHALLENGES AND HOW WE HANDLE THEM

Challenge / risk	Strategy
Real reaction colours differ from the printed chart	Validation by NCB/CFSL chemists under their licences: 1,296 captures at 3 sample amounts; we never handle narcotics
Phones and light vary; glare; night raids	Reference card + RAW capture; bad-light gate; flash/no-flash night mode; say "inconclusive" rather than guess
Staged or swapped sample	Capture starts before the reagent drop; liveness check; package label in frame; link to the s.105 video
Rooted phone, insider, deleted records	Hardware keys + attestation; supervisor-signed binding; panchnama anchor; server never trusted
Access to SIMS / e-Sakshya not confirmed	File exports first; the engine ships as an SDK e-Sakshya can embed

WHY IT IS VIABLE

- **Existing kit, existing phones:** NCB already supplies the kit free; the only addition is a printed colour card per kit.
- **Small infrastructure:** one app server pair, PostgreSQL, 1–3 TB a year of storage on NIC or MeitY-empanelled cloud.
- **Government-owned:** NCB holds the signing keys, the colour standard and the code; open formats, no licences.
- **Pass criteria set in advance:** false positives at most 1%, proven at 95% confidence (at least 299 negative controls); "inconclusive" at most 15%.
- **Clear status:** every number here is labelled proven in code or simulated. Real-phone and real-kit numbers come next.

WHO BENEFITS

- SO

Seizing officer
A guided test, a Field Test Memo and Form-1 item 5 filled in, and a record that shows good-faith action.
- IO

Investigating officer, prosecutor
Checkable evidence for remand and bail; a Rule 10(2) grouping check before sampling; the lab result linked back.
- CT

Courts, defence, the accused
Any expert can re-check the record offline; the quantity category is flagged as uncertain until the lab confirms.
- HQ

NCB headquarters and labs
India's first field-vs-lab accuracy data, per kit batch and per test.

WHAT CHANGES FOR ONE SEIZURE

Today (paper)	With FieldTest Recorder
Colour judged by eye	Measured against a signed colour standard
"Dark brown" written in the panchnama	Photo + reading + every drug the colour is consistent with
No proof of time or place	Signed time, GNSS time, GPS and officer
Records can be rewritten later	Deletions and re-runs are detectable
Quantity category guessed	Flagged when it can't be told in the field
No kit accuracy data	Field result vs lab result, per kit batch

Why a field report must not overstate: a real case

Anuraj v. State of Kerala (2024): police alleged commercial-quantity MDMA; the lab found methamphetamine in intermediate quantity (MDMA 0.5 g / 10 g vs methamphetamine 2 g / 50 g). Our record states only what the kit's colour supports, lists every drug that colour fits, and warns when they fall under different NDPS thresholds.

Social

Fairer remand and bail decisions; fewer cases lost on procedure; honest officers protected.

Economic

No new hardware; reuses the free NCB kit and officers' phones; small hosting cost.

Environmental

Paperless records; no extra consumables beyond one printed card per kit.

Scale

1,15,236 NDPS cases in 2022 (MHA); every package in every case gets a field test.

[1] **SIH26231 problem statement** — Narcotics Control Bureau, MHA; sih.gov.in

[2] **NCB Narcotic Drugs Detection Kit** — NICFS, A Forensic Guide for Crime Investigators, ch. 8, Figs 8.12–8.16 (Tests A–E, flow charts)

[3] **NDPS (Seizure, Storage, Sampling and Disposal) Rules 2022** — GSR 899(E): Rules 3, 5, 10(2), 13, 14

[4] **Sagar Parshuram Joshi v. State of Maharashtra** — Bombay HC, 15 Jan 2021 (indiankanoon.org/doc/90970274)

[5] **Masibur Khan v. State** — Delhi HC, 31 May 2023 (indiankanoon.org/doc/91398841)

[6] **Anuraj v. State of Kerala** — Kerala HC, 2024: alleged MDMA found to be methamphetamine

[7] **NDPS small and commercial quantities** — S.O. 1055(E), 19 Oct 2001

[8] **NCB Drug Law Enforcement Field Officers' Handbook** — narcoticsindia.nic.in

[9] **BPR&D SOP: audio-video recording under BNSS; BSA 2023 s.63** — hash value in the electronic-record certificate

[10] **NIJ Standard-0604.01** — Color Test Reagents/Kits for Preliminary Identification of Drugs of Abuse (US DoJ)

[11] **Android key attestation; RFC 8785 (JSON canonicalisation)** — [developer.android.com; rfc-editor.org](https://developer.android.com/rfc-editor.org)

[12] **MHA Rajya Sabha USQ 1832 (11 Dec 2024)** — NDPS case statistics

OUR OWN RESEARCH (AVAILABLE ON REQUEST)

- **Executable reference implementation** of the record format, verifier, colour engine, NCB kit protocol, searchable log and sync server; the Android app must match its outputs byte for byte.
- **17 experiments with 95% confidence intervals**, including 18 tampering attacks, 15 failure scenarios, sample-amount tests, night light and staged-sample detection.
- **An external review, re-checked claim by claim:** every confirmed defect was fixed and re-tested before this submission.
- **Safety and law:** our team never handles narcotics; colour tests use published colours, the kit chart and safe dyes. Real-kit validation is designed for NCB or CFSL chemists.